

ERIE, PA, USA

WMO: 725260

Lat: 42.080N

Lon: 80.182W

Elev: 729

StdP: 14.31

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14860

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	9.7	-3.9	4.6	6.3	0.8	5.9	11.5	27.5	37.8	25.9	33.0	9.8	190	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	86.9	73.0	84.4	71.8	82.1	70.6	75.3	83.2	73.8	81.3	72.5	79.9	10.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.6	124.3	80.6	71.2	118.1	79.0	69.8	112.4	77.7	39.2	83.4	37.8	81.4	36.5	79.9	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.4	21.1	19.2	DB	0.3	91.6	8.3	1.8	-5.7	93.0	-10.5	94.1	-15.2	95.1	-21.2	96.4	
			WB	-0.8	77.7	7.9	1.5	-6.4	78.8	-11.0	79.7	-15.4	80.5	-21.1	81.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	27.4	28.5	35.6	46.9	58.3	67.8	72.1	71.1	64.9	54.1	43.1	33.7
	DBStd	18.17	11.27	11.22	10.98	9.49	8.88	6.96	5.30	5.02	6.94	8.13	8.82	9.07
	HDD50	2773	704	606	468	172	21	0	0	0	1	47	244	510
	HDD65	6021	1165	1022	912	548	249	49	6	8	85	352	656	969
	CDD50	2925	4	5	22	79	278	534	685	653	448	173	38	6
	CDD65	700	0	0	1	5	41	134	226	196	83	14	0	0
	CDH74	4557	0	0	5	49	286	908	1587	1173	488	61	0	0
	CDH80	992	0	0	0	5	53	225	361	236	105	7	0	0
Wind	WSAvg	9.6	11.6	10.5	10.2	10.0	8.8	8.1	7.6	7.6	8.4	9.9	11.0	11.2
Precipitation	PrecAvg	43.1	2.8	2.4	3.0	3.5	3.5	4.1	3.6	4.0	4.6	3.8	4.1	3.9
	PrecMax	64.1	5.4	5.8	6.8	7.7	6.4	8.7	9.0	11.8	10.6	8.3	10.4	6.9
	PrecMin	31.1	1.0	0.6	0.7	1.6	1.0	0.7	0.5	0.5	1.3	1.3	1.5	1.7
	PrecStd	6.3	1.1	1.2	1.3	1.4	1.4	1.9	1.7	2.2	2.3	1.4	1.7	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.7	64.4	72.6	80.0	85.4	88.8	89.8	88.6	87.9	80.1	69.3	63.8
		MCWB	56.4	55.3	59.5	63.1	70.1	73.1	74.5	74.2	72.9	67.6	58.9	56.4
	2%	DB	55.9	56.1	64.9	73.6	80.9	85.4	86.5	85.4	83.0	74.8	64.5	56.5
		MCWB	51.7	50.0	54.6	59.6	67.5	71.7	73.5	72.8	70.7	64.3	56.1	50.3
	5%	DB	49.0	49.6	59.0	68.2	77.1	82.5	84.0	82.5	79.4	70.7	61.0	51.5
		MCWB	44.1	43.5	51.3	56.2	64.9	70.2	71.9	71.5	69.0	62.2	54.1	47.2
	10%	DB	42.9	43.6	52.7	63.0	73.0	79.5	81.5	80.0	76.3	66.6	57.0	46.2
		MCWB	39.1	38.8	45.6	53.4	63.0	68.8	70.9	70.3	67.4	59.0	50.3	42.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.3	57.0	60.4	65.1	72.0	76.0	77.5	77.0	75.3	70.2	61.4	57.4
		MCDB	61.2	63.7	70.4	76.4	82.3	84.5	86.3	85.2	84.5	77.6	66.4	61.7
	2%	WB	52.1	50.1	56.5	61.4	69.2	73.5	75.4	74.9	72.7	66.7	57.9	52.1
		MCDB	55.5	56.1	63.0	71.0	78.5	82.6	83.4	81.9	80.1	72.8	63.0	55.9
	5%	WB	44.8	44.0	51.4	58.2	66.7	71.7	73.9	73.3	70.6	63.3	54.6	46.8
		MCDB	48.6	48.6	57.9	66.4	75.0	80.5	81.4	79.9	77.5	69.0	60.1	50.6
	10%	WB	39.2	38.5	45.9	54.4	64.2	70.0	72.4	71.8	68.5	60.1	51.0	42.3
		MCDB	42.7	43.1	52.0	62.3	71.8	77.5	79.6	78.4	74.8	65.6	56.3	46.0
Mean Daily Temperature Range	5% DB	MDBR	12.2	13.3	14.2	16.7	16.5	15.3	14.6	14.4	14.7	13.7	11.9	10.4
		MCDBR	18.6	20.9	23.0	25.1	20.4	18.0	16.8	16.7	17.7	18.2	17.6	17.1
		MCWBR	16.0	16.6	15.5	15.3	11.3	9.6	9.0	9.3	9.8	10.8	13.1	14.5
	5% WB	MCDBR	18.1	20.0	21.8	23.3	18.7	15.8	14.0	14.2	15.3	15.7	16.6	16.7
		MCWBR	16.5	17.0	16.1	16.1	11.4	9.3	8.4	9.2	9.3	10.6	13.6	15.1
Clear-Sky Solar Irradiance	taub	0.304	0.341	0.343	0.380	0.414	0.440	0.438	0.424	0.384	0.361	0.316	0.299	
	taud	2.406	2.245	2.330	2.319	2.263	2.230	2.261	2.291	2.373	2.421	2.519	2.496	
	Ebn at Noon	267	270	286	282	274	266	265	265	269	262	262	260	
	Edh at Noon	25	35	36	39	42	44	42	40	34	29	23	21	
All-Sky Solar Radiation	RadAvg	450	682	1032	1414	1721	1897	1902	1683	1333	832	524	322	
	RadStd	36	82	125	148	163	142	141	90	127	81	76	33	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+187	N/A
Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	N/A

Nomenclature: See separate page